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TREND-BASED MOCK PAPER

GATE Electrical Engineering

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Electrical Engineering
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Choose synonym of 'pragmatic':

(Numerical Answer Type — Enter the numerical value)

Q2. [1-MCQ] [1 mark]

Speed = Distance /:

- (A) Time
- (B) Velocity
- (C) Acceleration
- (D) Force

Q3. [1-MCQ] [1 mark]

Distance 200km at 50km/h takes:

- (A) 2h
- (B) 4h
- (C) 5h
- (D) 3h

Q4. [1-MCQ] [1 mark]

Choose opposite of 'latent':

- (A) Obvious
 - (B) Hidden
 - (C) Clear
 - (D) Manifest
-

Section B: Electrical Engineering

Q5. [1-MCQ] [1 mark]

Full-wave rectifier uses:

- (A) 1
- (B) 2
- (C) 4
- (D) 3

Q6. [1-MCQ] [1 mark]

Step-down chopper output:

- (A) Less than input
- (B) More than input
- (C) Equal
- (D) Zero

Q7. [1-MCQ] [1 mark]

Inverter converts:

- (A) DC to AC
- (B) AC to DC
- (C) DC to DC
- (D) AC to AC

Q8. [1-MCQ] [1 mark]

Thyristor is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q9. [2-NAT] [2 marks]

Full-wave rectifier PIV across each diode:

(Numerical Answer Type — Enter the numerical value)

Q10. [2-MCQ] [2 marks]

Step-down chopper $V_s=200V$, $\alpha=0.6$. Output:

- (A) 120V
- (B) 80V
- (C) 200V
- (D) 333V

Q11. [2-MCQ] [2 marks]

Inductor current in CCM of DC-DC converter:

- (A) Always zero
- (B) Never zero
- (C) Always constant
- (D) Oscillating

Q12. [2-NAT] [2 marks]

For a buck converter, $V_{out} =$

(Numerical Answer Type — Enter the numerical value)

Q13. [2-NAT] [2 marks]

IGBT combines features of:

(Numerical Answer Type — Enter the numerical value)

Q14. [2-MCQ] [2 marks]

3-phase inverter 120-degree conduction eliminates:

- (A) 3rd harmonic
 - (B) 5th harmonic
 - (C) 7th harmonic
 - (D) No triplen
-

Q15. [1-NAT] [1 mark]

Integral controller eliminates:

(Numerical Answer Type — Enter the numerical value)

Q16. [1-MCQ] [1 mark]

Root locus plots:

- (A) Open loop poles
 - (B) Closed loop poles
 - (C) Both
 - (D) None
-

Q17. [1-MCQ] [1 mark]

Gain margin:

- (A) At phase crossover
 - (B) At gain crossover
 - (C) Both
 - (D) None
-

Q18. [1-MCQ] [1 mark]

Time constant of RC:

- (A) RC
 - (B) R/C
 - (C) C/R
 - (D) 1/RC
-

Q19. [2-MCQ] [2 marks]

Gain margin is defined at:

- (A) Phase crossover
 - (B) Gain crossover
 - (C) Both
 - (D) Neither
-

Q20. [2-MSQ] [2 marks]

Integral controller eliminates:

- (A) Rise time
- (B) Steady-state error
- (C) Damping
- (D) Overshoot

Q21. [2-NAT] [2 marks]

Root locus branches start at poles and end at:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-NAT] [2 marks]

A system with characteristic $1+s = 0$ has:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

Nyquist criterion: $N = Z - P$. If $P=1$, $N=-1$:

- (A) 1 zero in RHP
 - (B) Stable
 - (C) 1 clockwise encirclement needed
 - (D) Unstable
-

Q24. [1-MCQ] [1 mark]

DC starter used to:

- (A) Regulate speed
 - (B) Limit start current
 - (C) Reverse
 - (D) Increase torque
-

Q25. [1-NAT] [1 mark]

Alternator produces:

(Numerical Answer Type — Enter the numerical value)

Q26. [1-MCQ] [1 mark]

Slip $s =$

- (A) $(N_s - N_r)/N_s$
 - (B) $(N_r - N_s)/N_s$
 - (C) N_s/N_r
 - (D) N_r/N_s
-

Q27. [1-NAT] [1 mark]

Transformer works on:

(Numerical Answer Type — Enter the numerical value)

Q28. [1-MCQ] [1 mark]

Slip in induction motor:

- (A) Always positive
 - (B) Can be negative
 - (C) Always zero
 - (D) Infinity
-

Q29. [2-MCQ] [2 marks]

DC shunt motor: $V=220V$, $R_{sh}=220\Omega$, $R_a=1\Omega$, $I=50A$. Back emf:

- (A) 171V
- (B) 219V
- (C) 220V
- (D) 225V

Q30. [2-MCQ] [2 marks]

Max torque in IM at start ($s=1$) requires:

- (A) $R_2 = X_2$
- (B) $R_2 = 0$
- (C) $X_2 = 0$
- (D) $R_2 \gg X_2$

Q31. [3-MCQ] [3 marks]

3-phase transformer primary:secondary turns = 10:1. If primary is delta, secondary is:

- (A) Delta
- (B) Star
- (C) Both
- (D) Either

Q32. [1-NAT] [1 mark]

Per unit system base values:

(Numerical Answer Type — Enter the numerical value)

Q33. [1-MCQ] [1 mark]

Synchronous generator excitation:

- (A) Rotor
- (B) Stator
- (C) Both
- (D) None

Q34. [1-MCQ] [1 mark]

Most efficient transmission:

- (A) Low voltage
- (B) High voltage
- (C) DC
- (D) AC

Q35. [1-MCQ] [1 mark]

Circuit breaker operates on:

- (A) Overload
- (B) Short circuit
- (C) Both
- (D) None

Q36. [2-MSQ] [2 marks]

Transmission line: sending end voltage higher than receiving for:

- (A) Heavy loading
- (B) Light loading (capacitive)
- (C) Unity PF
- (D) No load

Q37. [2-NAT] [2 marks]

3-phase, 4-pole IM at 1440 rpm on 50 Hz. Slip:

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MCQ] [2 marks]

Swing equation relates:

- (A) Power and voltage
- (B) Accelerating power and rotor angle
- (C) Flux and current
- (D) Torque and speed

Q39. [2-NAT] [2 marks]

Per unit reactance: $X_{\text{ohm}}=10$, base=100 MVA, kV=132. $X_{\text{pu}} =$

(Numerical Answer Type — Enter the numerical value)

Q40. [1-MCQ] [1 mark]

Eigenvalues of identity:

- (A) All zero
- (B) All one
- (C) Distinct
- (D) Depends

Q41. [2-NAT] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q43. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q44. [2-NAT] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q45. [2-NAT] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q46. [2-NAT] [2 marks]

Superposition theorem applies to:

(Numerical Answer Type — Enter the numerical value)

Q47. [2-MCQ] [2 marks]

Delta-Star transformation: $R_a(\text{star}) =$

- (A) $R1 \cdot R2 / (R1 + R2 + R3)$
- (B) $(R1 \cdot R2 + R2 \cdot R3 + R3 \cdot R1) / R3$
- (C) $R1 + R2$
- (D) $R1 \cdot R2 / R3$

Q48. [2-NAT] [2 marks]

Norton equivalent of Thevenin ($V_{th}=10V$, $R_{th}=5\Omega$):

(Numerical Answer Type — Enter the numerical value)

Q49. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q50. [2-MCQ] [2 marks]

Y-parameter Y_{11} for series impedance Z is:

- (A) $1/Z$
- (B) Z
- (C) $-Z$
- (D) 0

Q51. [1-NAT] [1 mark]

LTI characterized by:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-MCQ] [2 marks]

Nyquist rate for 4kHz signal:

- (A) 4kHz
- (B) 8kHz
- (C) 2kHz
- (D) 1kHz

Q53. [2-MCQ] [2 marks]

Aliasing avoided by:

- (A) Increasing rate
- (B) Decreasing rate
- (C) Both
- (D) None

Q54. [1-MCQ] [1 mark]

Negative feedback:

- (A) Increases gain
- (B) Decreases gain
- (C) No effect
- (D) Oscillates

Q55. [1-MCQ] [1 mark]

Ideal op-amp gain:

- (A) Infinite
- (B) Zero
- (C) Unity
- (D) Negative

Q56. [1-NAT] [1 mark]

RC phase shift uses:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

ICBO in transistor is:

- (A) Collector-Base reverse
- (B) Emitter-Base
- (C) Collector-Emitter
- (D) None

Q58. [1-NAT] [1 mark]

Universal gate:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

JK flip-flop toggles when:

- (A) $J=K=0$
- (B) $J=K=1$
- (C) $J=0, K=1$
- (D) $J=1, K=0$

Q60. [1-MCQ] [1 mark]

Mod-10 counts from:

- (A) 0 to 10
- (B) 0 to 9
- (C) 1 to 10
- (D) 1 to 9

Q61. [1-MCQ] [1 mark]

PLD stands for:

- (A) Programmable Logic Device
- (B) Processor Logic
- (C) Programmable Logic Design
- (D) None

Q62. [2-MCQ] [2 marks]

Corona loss increases with:

- (A) Wire diameter
- (B) Voltage
- (C) Both
- (D) None

Q63. [2-NAT] [2 marks]

Displacement current added by:

(Numerical Answer Type — Enter the numerical value)

Q64. [2-NAT] [2 marks]

Compound interest: P at r% for 2 years =

(Numerical Answer Type — Enter the numerical value)

Q65. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
 - (B) 27
 - (C) 9
 - (D) 3
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	A	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1MCQ	B	1
6	1MCQ	B	1
7	1NAT	B	1
8	1MCQ	A	1
9	1NAT	B	1
10	1MCQ	A	1
11	1NAT	C	1
12	1MCQ	A	1
13	1MCQ	B	1

14	1MCQ	C	1
15	1NAT	B	1
16	1MCQ	B	1
17	1MCQ	A	1
18	1MCQ	A	1
19	1MCQ	C	1
20	1MCQ	A	1
21	1MCQ	A	1
22	1MCQ	A	1
23	1MCQ	B	1
24	1MCQ	A	1
25	1NAT	B	1
26	1MCQ	A	1
27	1NAT	C	1
28	1MCQ	B	1

29	1MCQ	B	1
30	1MCQ	A	1
31	1NAT	D	1
32	2MCQ	B	2
33	2MCQ	A	2
34	2MCQ	C	2
35	2NAT	B	2
36	2MCQ	A	2
37	2NAT	B	2
38	2MSQ	B	2
39	2NAT	B	2
40	2MCQ	A	2
41	2NAT	A	2
42	2MCQ	A	2
43	2MCQ	A	2

44	2MSQ	B	2
45	2NAT	A	2
46	2MCQ	A	2
47	2NAT	A	2
48	2MCQ	B	2
49	2MCQ	B	2
50	2MCQ	B	2
51	2NAT	C	2
52	2NAT	B	2
53	2NAT	A	2
54	2NAT	A	2
55	2MCQ	C	2
56	2NAT	A	2
57	2MCQ	B	2
58	2MCQ	A	2

59	3MCQ	B	3
60	2NAT	A	2
61	2NAT	A	2
62	2MCQ	D	2
63	2NAT	A	2
64	2MCQ	A	2
65	2MCQ	D	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: DC Machines | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: Alternators | Difficulty: difficult

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Electrical Machines | Topic: Induction Motor | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: Transformers | Difficulty: difficult

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Electrical Machines | Topic: Induction Motor | Difficulty: moderate

Q11.

Answer: C

Solution:

Correct Answer: C

Subject: Power Systems | Topic: Per Unit | Difficulty: easy

Q12.

Answer: A

Solution:

Correct Answer: A

Subject: Power Systems | Topic: Generators | Difficulty: easy

Q13.

Answer: B

Solution:

Correct Answer: B

Subject: Power Systems | Topic: Transmission | Difficulty: moderate

Q14.

Answer: C

Solution:

Correct Answer: C

Subject: Power Systems | Topic: Protection | Difficulty: difficult

Q15.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Controllers | Difficulty: moderate

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Root Locus | Difficulty: moderate

Q17.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Frequency Response | Difficulty: easy

Q18.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Time Response | Difficulty: easy

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Power Electronics | Topic: Rectifiers | Difficulty: moderate

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Choppers | Difficulty: moderate

Q21.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Inverters | Difficulty: difficult

Q22.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Converters | Difficulty: moderate

Q23.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Electronics | Topic: Feedback | Difficulty: moderate

Q24.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Electronics | Topic: Op-Amps | Difficulty: moderate

Q25.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Electronics | Topic: Oscillators | Difficulty: easy

Q26.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Electronics | Topic: Transistors | Difficulty: moderate

Q27.

Answer: C

Solution:

Correct Answer: C

Subject: Digital Electronics | Topic: Logic Gates | Difficulty: moderate

Q28.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Electronics | Topic: Flip-Flops | Difficulty: easy

Q29.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Electronics | Topic: Counters | Difficulty: easy

Q30.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Electronics | Topic: PLD | Difficulty: moderate

Q31.

Answer: D

Solution:

Correct Answer: D

Subject: Signals and Systems | Topic: LTI Systems | Difficulty: moderate

Q32.

Answer: B

Solution:

Correct Answer: B

Subject: Signals and Systems | Topic: Sampling | Difficulty: moderate

Q33.

Answer: A

Solution:

Correct Answer: A

Subject: Signals and Systems | Topic: Sampling | Difficulty: moderate

Q34.

Answer: C

Solution:

Correct Answer: C

Subject: EMFT | Topic: Transmission | Difficulty: difficult

Q35.

Answer: B

Solution:

Correct Answer: B

Subject: EMFT | Topic: Maxwell | Difficulty: moderate

Q36.

Answer: A

Solution:

$I_{sh} = 220/220 = 1A$. $I_a = 49A$. $E_b = 220 - 49 \times 1 = 171V$.

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: moderate

Q37.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: B

Solution:

Ferranti effect: light load/capacitive charging causes $V_s > V_r$. Long lines at no-load.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q39.

Answer: B

Solution:

$N_{sync} = 120 \times 50/4 = 1500$. $s = (1500 - 1440)/1500 = 60/1500 = 0.04 = 4\%$.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q40.

Answer: A

Solution:

Gain margin: at phase crossover frequency (phase = -180 deg), $1/|GH(j\omega)|$ in dB.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q41.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Measurements | Topic: Measurements | Difficulty: moderate

Q42.

Answer: A

Solution:

Max torque at $s = R_2/X_2$. For start ($s=1$): $R_2 = X_2$ gives max starting torque.

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: easy

Q43.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3 + 1/x^3 = 27 - 3 \times 3 = 18$.

Subject: Measurements | Topic: Measurements | Difficulty: moderate

Q44.

Answer: B

Solution:

Integral control: eliminates steady-state error for step/ramp inputs. Increases system type by 1.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q45.

Answer: A

Solution:

Full-wave: reverse voltage across non-conducting diode = V_m .

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q46.

Answer: A

Solution:

$V_o = \alpha * V_s = 0.6 * 200 = 120V$.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q47.

Answer: A

Solution:

Superposition requires linearity (homogeneity + additivity). Does NOT apply to non-linear.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q48.

Answer: B

Solution:

$R_a = (R_1 * R_2 + R_2 * R_3 + R_3 * R_1) / R_3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q49.

Answer: B

Solution:

Continuous conduction mode (CCM): inductor current never reaches zero.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: easy

Q50.

Answer: B

Solution:

$M * d^2(\delta) / dt^2 = P_m - P_e$. Relates accelerating power to rotor angle dynamics.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q51.

Answer: C

Solution:

Root locus: starts at OL poles, ends at OL zeros (or infinity if $P > Z$).

Subject: Control Systems | Topic: Control Systems | Difficulty: easy

Q52.

Answer: B

Solution:

Root at $s = -1$ (on left half plane). Simple pole on negative real axis: stable (marginally, as $t \rightarrow \infty$).

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q53.

Answer: A

Solution:

Buck (step-down): $V_o = D * V_{in}$ where D is duty cycle.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q54.

Answer: A

Solution:

$I_n = V_{th} / R_{th} = 10/5 = 2A$. $R_n = R_{th} = 5\Omega$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q55.

Answer: C

Solution:

$N = Z - P$. $N = -1$ means one clockwise encirclement of -1 point needed. $Z = N + P = -1 + 1 = 0$. Stable.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q56.

Answer: A

Solution:

$X_{pu} = X * MVA_{base} / (kV_{base})^2 = 10 * 100 / 132^2 = 1000/17424 = 0.0574$. Hmm. With formula $X_{pu} = X * (MVA_{base} / (kV_{base}^2)) * 10 = 10 * 100 / (132^2) * 10 = 10000/17424 = 0.574$. Option A.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q57.

Answer: B

Solution:

$proj = ((v.w)/(w.w))*w = (4/5)*(2,1,0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q58.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: B

Solution:

Dy connection: primary delta, secondary star. Common for step-down (reduces voltage, provides neutral).

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: easy

Q60.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: A

Solution:

IGBT: BJT (high current) + MOSFET (easy drive). High input impedance, low saturation voltage.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q62.

Answer: D

Solution:

120-degree conduction: triplen harmonics (3rd, 9th, ...) eliminated in line voltage.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q63.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: A

Solution:

$L\{e^{-(at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q65.

Answer: D

Solution:

Series Z: $I_1=I_2$, $V_1-V_2=Z*I_1$. $Y_{11} = I_1/V_1$ with $I_2=0$ impossible ($I_1=I_2$). $Y_{11}=0$ for pure series.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Power Electronics: 10 questions, 16 marks
Control Systems: 9 questions, 14 marks
Electrical Machines: 8 questions, 12 marks
Power Systems: 8 questions, 12 marks
Engineering Mathematics: 6 questions, 11 marks
Network Theory: 5 questions, 10 marks
Signals and Systems: 3 questions, 5 marks
General Aptitude: 4 questions, 4 marks
Analog Electronics: 4 questions, 4 marks
Digital Electronics: 4 questions, 4 marks
EMFT: 2 questions, 4 marks
Measurements: 2 questions, 4 marks

Difficulty Distribution:

Easy: 15 | Moderate: 44 | Difficult: 6

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.